

1. Project Name

Dialysis - A DTMF Signal Analysis Studio

2. Description

An interactive desktop application built with Python and PyQt for the synthesis, visualization, and decoding of **Dual-Tone Multi-Frequency (DTMF)** signals. The platform provides an end-to-end transmit and receive signal processing pipeline. Users can generate single or multi-digit tone sequences through a virtual keypad, export custom tone sequences to WAV audio files, and upload external audio recordings for automated digit sequence detection. In addition, the studio provides real-time time-domain waveforms, magnitude spectra, dynamic spectrograms, pole-zero Z-transform designs, discrete-time LTI system visualizers, and performance benchmarking under noise degradation and variable sampling rates.

3. Key Features

- **Virtual Keypad & Tone Synthesis:**

- Interactive 4×4 keypad generating composite DTMF sinusoids $x[n] = \sin(2\pi f_1 n / f_s) + \sin(2\pi f_2 n / f_s)$ across low (697–941 Hz) and high (1209–1633 Hz) frequency groups.
- Keyboard sequence typing support with instant real-time audio tone playback.

- **Audio File Exporter:**

- Exports user-entered multi-digit keypad strings into uncompressed WAV audio files.

- **Audio File Sequence Detector:**

- Accepts user-uploaded audio files, automatically segments active tone regions, and extracts the full embedded digit sequence.

- **Spectral & Time-Domain Inspector:**

- Displays time-domain waveforms, dynamic spectrograms, and magnitude spectra with dual-tone frequency peak highlights.

- **Goertzel Algorithm Decoder:**

- Decodes dual tones via a 2nd-order recursive LTI difference equation designed directly in the Z-domain with poles centered on target frequencies.

- **Fast Fourier Transform Decoder:**

- Parallel decoding pipeline utilizing FFT spectral peak detection across low and high frequency bands.

- **Pole-Zero Z-Transform Visualizer:**

- Interactive Z-plane plot displaying filter pole-zero locations, stability boundaries, and resonator selectivity.

- **Additive Noise Injector:**

- Real-time AWGN noise control slider to evaluate decoding accuracy degradation under low signal-to-noise ratios.

- **Variable Sampling Rate Controller:**

- Adjustable sampling rate f_s slider to observe spectral aliasing and demonstrate the Nyquist-Shannon sampling theorem.

- **LTI Impulse Response & Convolution Visualizer:**

- Renders filter impulse response $h[n]$ and illustrates time-domain discrete convolution $y[n] = x[n] * h[n]$ to explain filter resonance during tone detection.

- **Performance Benchmark Analyzer:**

- Side-by-side comparative analysis of **FFT vs. Goertzel** decoders measuring execution speed, memory consumption, and accuracy.